

Data transformations in Power BI

# Data cleaning and formatting in Power BI

# Data cleaning and formatting

Data cleaning and formatting are **critical steps** in creating visualisations and making data-driven decisions.

Clean data ensures the **accuracy** and **reliability** of insights derived from visualisations, preventing misinformed conclusions due to errors or inconsistencies.

Clean data enhances the **interpretability** of visualisations, making it easier for stakeholders to grasp and act upon the information presented, ultimately facilitating better decision-making.

# Naming conventions

User-friendly naming conventions **enhance the readability of a data model**. There are several ways to rename columns and queries in the Power Query Editor. Using the **"Transform"** tab is one method.

01.

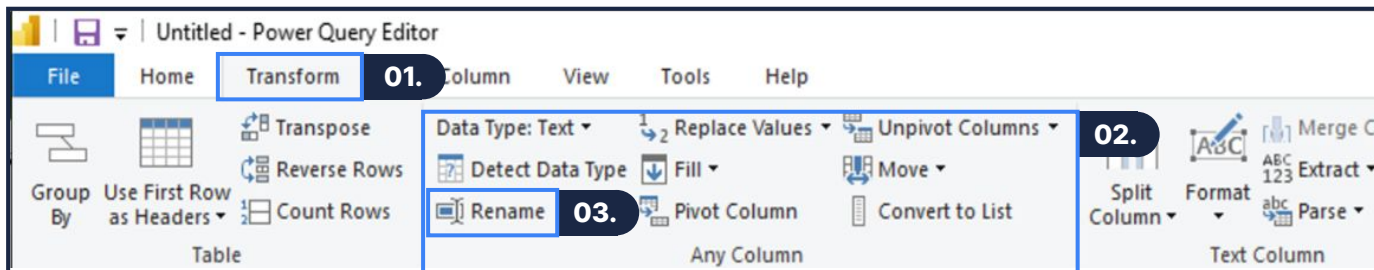
Click on the **"Transform"** tab.

02.

Navigate to the **"Any column"** section which contains transformations at the column level for any data type column.

03.

Select **"Rename"**.

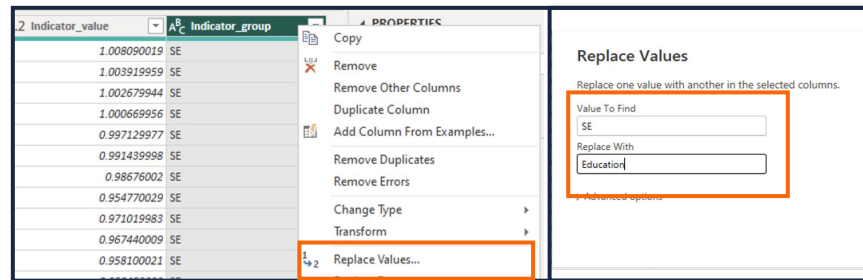


# Resolve data quality issues

Power Query Editor enables us to address various data quality issues, such as inconsistencies, unexpected values, and null values by either **replacing** or **deleting** them.

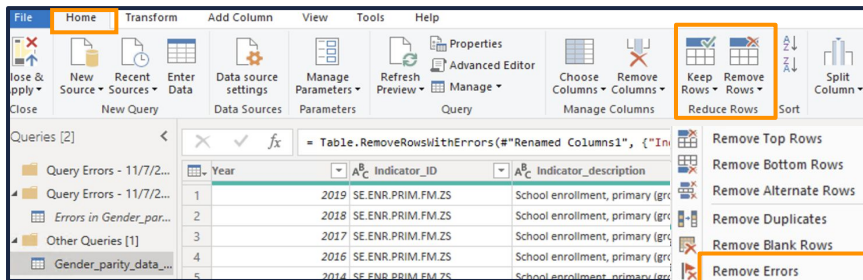
## Replace values

- Right-click on the column name.
- Select **"Replace Values"**.
- Enter the **"Value To Find"** and the value to **"Replace With"**.



## Reduce rows

- Select the column name.
- Navigate to the **"Home"** tab.
- Select **"Remove Rows"** under the **"Reduce Rows"** section.
- Select **"Remove Errors"** in the dropdown menu.

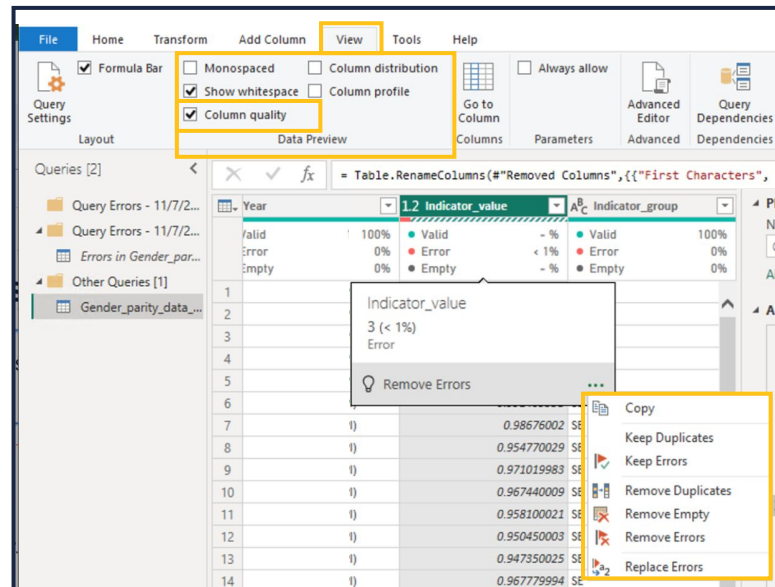


# Resolve data quality issues

The "**View**" tab in Power Query Editor also allows us to see the **column quality** of each column in the dataset and resolve any data quality issues.

## Column quality view

- Right-click on the column name.
- Navigate to the "**View**" tab.
- Select the "**Column quality**" check box in the "**Data Preview**" section.
- Select the appropriate data quality solution.

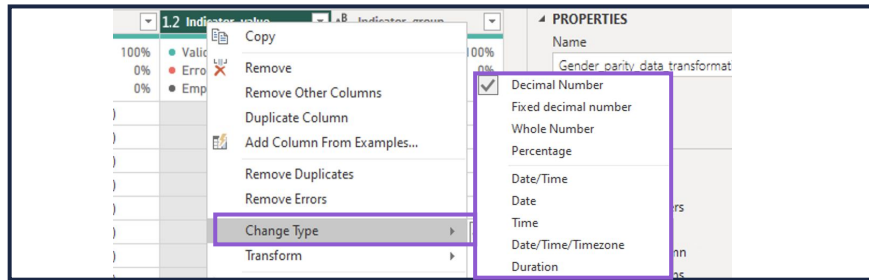


# Evaluate and transform data types

We can **evaluate and change data types** to ensure data consistency. In the Power Query Editor, there are multiple ways to transform data types.

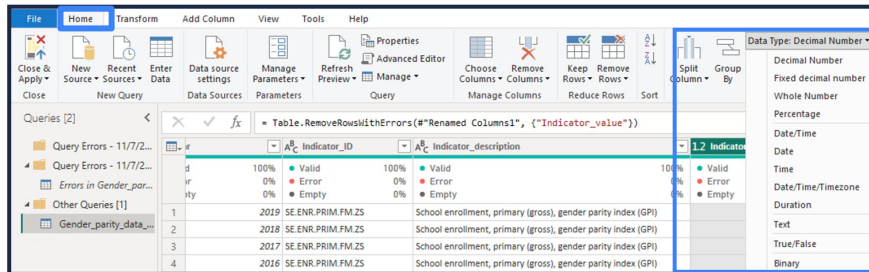
## Right-click method

- Right-click on the column name.
- Select **“Change Type”**.
- Select the appropriate data type from the dropdown menu.



## Transform section method

- Click on the **“Home”** tab.
- Navigate to the **“Transform”** section, which contains general transformations at the column level.
- Select the appropriate data type from the dropdown menu.

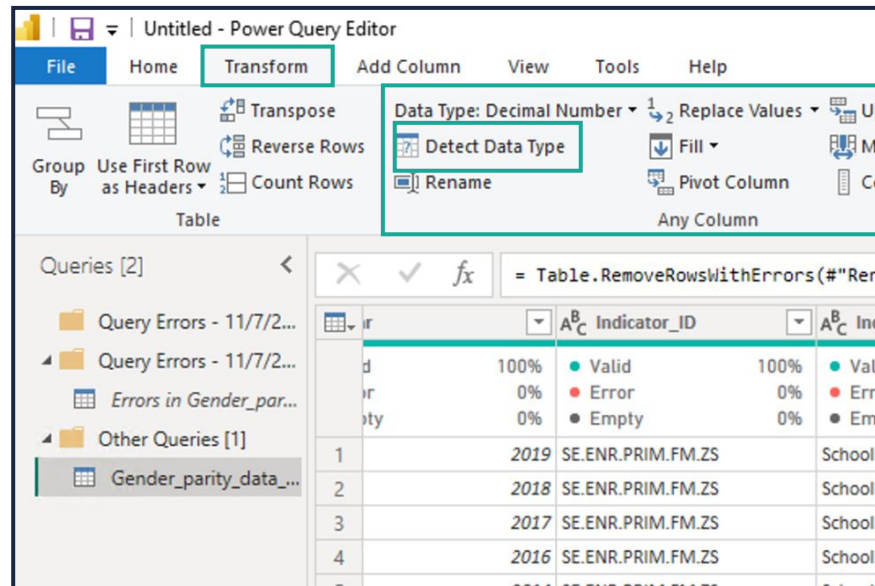


# Evaluate and transform data types

We can also use the Power Query Editor to identify data types automatically based on the values in the column.

## Detect the data type

- Select the relevant column.
- Navigate to the **“Transform”** tab.
- Select the **“Detect Data Type”** option in the **“Any Column”** section.



The screenshot shows the Power Query Editor interface. The 'Transform' tab is selected in the ribbon. In the 'Any Column' section, the 'Detect Data Type' option is highlighted with a red box. The 'Data Type: Decimal Number' dropdown is also visible. Below the ribbon, the 'Queries' list on the left shows 'Gender\_parity\_data\_...' selected. The main area displays a table with columns 'Indicator\_ID' and 'Indicator\_Name'. The table contains data for years 2019, 2018, 2017, and 2016, all with the value 'SE.ENR.PRIM.FM.ZS' and 'School' as the indicator name.

|   | Indicator_ID | Indicator_Name    |
|---|--------------|-------------------|
| 1 | 2019         | SE.ENR.PRIM.FM.ZS |
| 2 | 2018         | SE.ENR.PRIM.FM.ZS |
| 3 | 2017         | SE.ENR.PRIM.FM.ZS |
| 4 | 2016         | SE.ENR.PRIM.FM.ZS |

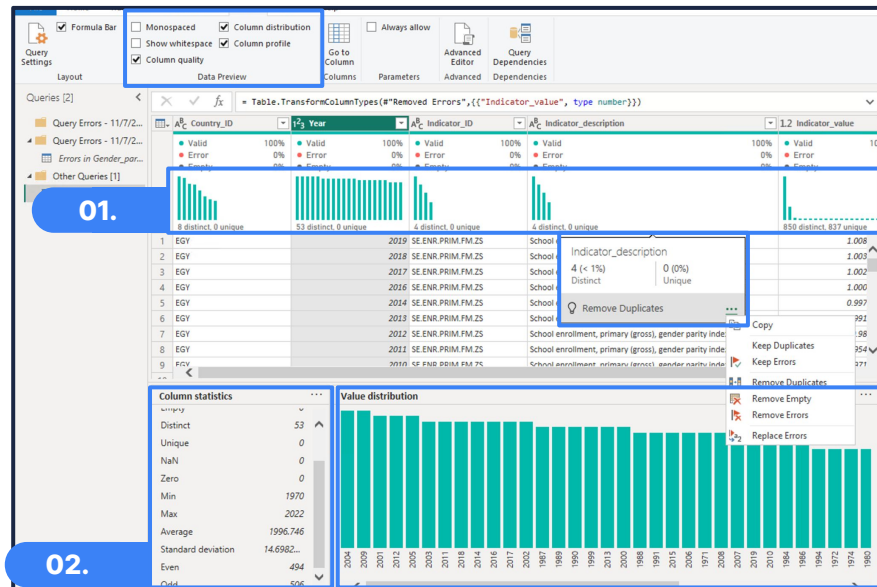
# Data profiling

Profiling data helps us understand its characteristics. We can use profiling to **identify outliers**, **missing values**, and **data distributions**. Power Query Editor provides summary statistics and histograms to help us achieve this.

Navigate to the “**View**” tab. Under the “**Data Preview**” section, select “**Column quality**”, “**Column distribution**”, and “**Column profile**”.

**01. Column distribution:** Displays a **simple distribution** of the data within the column, the **number of distinct column values**, and how many of those are **unique**. Running our mouse over this graphic will provide us with some options for eliminating duplicate values.

**02. Column profile:** Based on the selected column, we may zoom into a single column and examine some column summary statistics such as **minimum**, **maximum**, **average**, **standard deviation**, etc. We can also see how the values in that column are **distributed**.





# Apply data shape transformations

Transforming data shapes involves grouping, pivoting, unpivoting, or transposing data. The Power Query Editor allows us to achieve this under the **"Transform"** tab.

01.

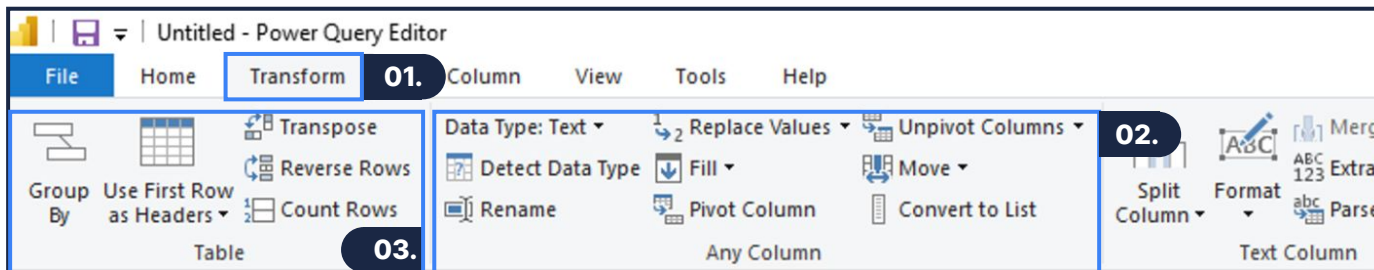
Navigate to the **"Transform"** tab.

02.

The **"Any Column"** feature allows us to pivot or unpivot columns to change the shape of our dataset as well as move columns around.

03.

The **"Table"** section allows us to group data, transpose and rearrange rows.



# Group by

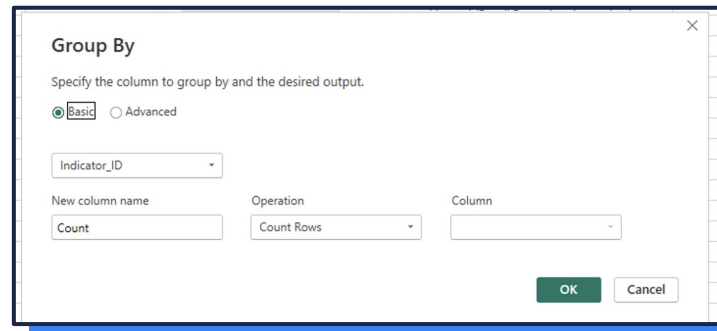
To group data by a specific column, we use the group by option which is either available in the “**Table**” section under the “**Transform**” tab or in the “**Home**” tab under the “**Transform**” section.

## Basic:

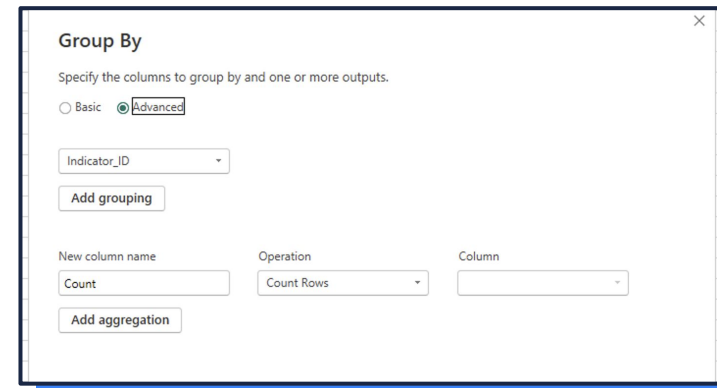
Under the basic settings, we can **select the column to group by**, as well as the **operation to use to group** the column and the name of the grouped column.

## Advanced:

Under the advanced settings, we can **select multiple columns to group by**. Selecting the “Add grouping” button will allow us to choose a new grouping column. We can also use **multiple aggregations for our groupings** by selecting the “Add aggregations” button.



The screenshot shows the 'Group By' dialog box with the 'Basic' tab selected. The title is 'Group By' and the instruction is 'Specify the column to group by and the desired output.' There are two radio buttons: 'Basic' (selected) and 'Advanced'. Below the radio buttons is a dropdown menu for 'Indicator\_ID'. Underneath, there are three fields: 'New column name' with a text input containing 'Count', 'Operation' with a dropdown menu showing 'Count Rows', and 'Column' with an empty dropdown menu. At the bottom right are 'OK' and 'Cancel' buttons.



The screenshot shows the 'Group By' dialog box with the 'Advanced' tab selected. The title is 'Group By' and the instruction is 'Specify the columns to group by and one or more outputs.' There are two radio buttons: 'Basic' and 'Advanced' (selected). Below the radio buttons is a dropdown menu for 'Indicator\_ID'. Underneath is an 'Add grouping' button. At the bottom, there are three fields: 'New column name' with a text input containing 'Count', 'Operation' with a dropdown menu showing 'Count Rows', and 'Column' with an empty dropdown menu. Below these fields is an 'Add aggregation' button.

# Transposing, pivoting, and unpivoting data

In the Power Query Editor, there are two distinct **data shape transformation operations**, namely **transposing** and **pivoting** data. Each operation serves a different purpose and can be used to restructure data in various ways.

## Transpose

Convert **columns to rows** and **rows to columns** for the entire table.

Allows us to switch from a **wide format** (many columns, few rows) to a **tall format** (few columns, many rows) or vice versa.

Select the table to transpose and click on **“Transpose”** under the **“Table”** section.

| Country_ID | Country_name | Major_region | Region          | Development            |
|------------|--------------|--------------|-----------------|------------------------|
| 1 EGY      | Egypt        | Africa       | Northern Africa | Less developed regions |
| 2 ETH      | Ethiopia     | Africa       | Eastern Africa  | Less developed regions |
| 3 GHA      | Ghana        | Africa       | Western Africa  | Less developed regions |
| 4 KEN      | Kenya        | Africa       | Eastern Africa  | Less developed regions |
| 5 MAR      | Morocco      | Africa       | Northern Africa | Less developed regions |
| 6 RWA      | Rwanda       | Africa       | Eastern Africa  | Less developed regions |
| 7 NGA      | Nigeria      | Africa       | Western Africa  | Less developed regions |
| 8 ZAF      | South Africa | Africa       | Southern Africa | Less developed regions |

| Column1                  | Column2                | Column3                | Column4                | Column5                | Column6                |
|--------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| 1 EGY                    | ETH                    | GHA                    | KEN                    | MAR                    | RWA                    |
| 2 Egypt                  | Ethiopia               | Ghana                  | Kenya                  | Morocco                | Rwanda                 |
| 3 Africa                 | Africa                 | Africa                 | Africa                 | Africa                 | Africa                 |
| 4 Northern Africa        | Eastern Africa         | Western Africa         | Eastern Africa         | Northern Africa        | Eastern Africa         |
| 5 Less developed regions | Less developed regions | Less developed regions | Less developed regions | Less developed regions | Less developed regions |

# Transposing, pivoting, and unpivoting data

## Pivot

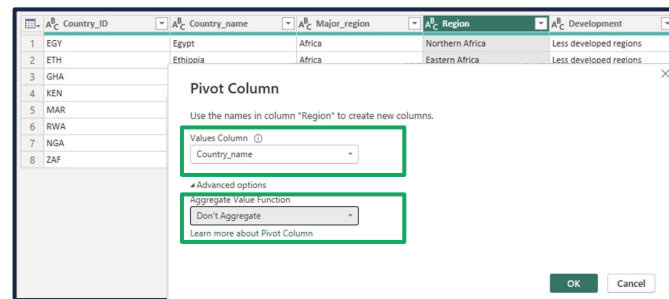
Convert rows into columns for a specific column.

Used to **aggregate data** and create summary tables. It converts unique values in a column into new columns, allowing us to **summarise data** by specific attributes.

Navigate to the "**Transform**" tab and select "**Pivot Column**" under the "**Any Column**" section.

**01.** Select where the **cell values for the new columns** will come from.

**02.** We can also specify an **aggregation function**, like sum or average, for the pivoted values.



| Major_region | Development            | Northern Africa | Eastern Africa | Western Africa |
|--------------|------------------------|-----------------|----------------|----------------|
| Africa       | Less developed regions | Egypt           | null           | null           |
| Africa       | Less developed regions | null            | Ethiopia       | null           |
| Africa       | Less developed regions | null            | null           | Ghana          |
| Africa       | Less developed regions | null            | null           | Kenya          |
| Africa       | Less developed regions | Morocco         | null           | null           |
| Africa       | Less developed regions | null            | null           | Nigeria        |
| Africa       | Less developed regions | null            | Rwanda         | null           |
| Africa       | Less developed regions | null            | null           | null           |